

LearnCore Database Schema DBML

```
Table users {
  id bigint [pk, increment]
  name varchar(255) [not null]
  email varchar(255) [not null, unique]
  email_verified_at timestamp
  email_otp_code varchar(10)
  email_otp_sent_at timestamp
  email_otp_expires_at timestamp
  email_otp_verified_at timestamp
  password varchar(255) [not null]
  remember_token varchar(100)
  created_at timestamp
  updated_at timestamp
}

Table faculties {
  faculty_id bigint [pk, increment]
  faculty_code varchar(30) [not null, unique]
  faculty_name varchar(150) [not null]
  created_at timestamp
  updated_at timestamp
}

Table departments {
  department_id bigint [pk, increment]
  faculty_id bigint [not null]
  department_code varchar(30) [not null, unique]
  department_name varchar(150) [not null]
  deans varchar(255)
  created_at timestamp
  updated_at timestamp
}

Table course_categories {
  course_category_id bigint [pk, increment]
  category_code varchar(30) [not null, unique]
  category_name varchar(150) [not null]
  description text
  created_at timestamp
  updated_at timestamp
}

Table academic_years {
  academic_year_id bigint [pk, increment]
  year_name varchar(50) [not null, unique]
  start_date date
  end_date date
  is_active boolean [not null, default: false]
  created_at timestamp
  updated_at timestamp
}

Table semesters {
  semester_id bigint [pk, increment]
  academic_year_id bigint [not null]
  semester_name varchar(100) [not null]
  start_date date [not null]
  end_date date [not null]
  is_active boolean [not null, default: false]
  created_at timestamp
  updated_at timestamp

  Indexes {
    (academic_year_id, semester_name) [unique]
  }
}

Table courses {
  course_id bigint [pk, increment]
  course_category_id bigint [not null]
  department_id bigint
  academic_year_id bigint
  semester_id bigint
  course_code varchar(30) [not null, unique]
  course_name varchar(150) [not null]
  description text
  created_at timestamp
  updated_at timestamp
}
```

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```
Indexes {
  (department_id, academic_year_id, semester_id) [name: 'courses_academic_scope_idx']
  (course_category_id, department_id) [name: 'courses_category_department_idx']
}

Table subjects {
  subject_id bigint [pk, increment]
  course_id bigint [not null]
  subject_code varchar(30) [not null, unique]
  subject_name varchar(150) [not null]
  credit smallint [not null, default: 0]
  description text
  created_at timestamp
  updated_at timestamp
}

Table class_rooms {
  class_room_id bigint [pk, increment]
  course_id bigint [not null]
  academic_year_id bigint
  class_code varchar(30) [not null, unique]
  class_name varchar(150) [not null]
  room varchar(100)
  created_at timestamp
  updated_at timestamp
}

Table students {
  student_id bigint [pk, increment]
  user_id bigint [unique]
  student_code varchar(30) [not null, unique]
  department_id bigint
  academic_year_id bigint
  semester_id bigint
  first_name varchar(100) [not null]
  last_name varchar(100)
  gender varchar(20)
  date_of_birth date
  phone varchar(30)
  email varchar(150) [unique]
  address text
  status varchar(30) [not null, default: 'active']
  created_at timestamp
  updated_at timestamp

  Indexes {
    (department_id, academic_year_id, semester_id, status) [name: 'students_academic_status_idx']
    (first_name, last_name) [name: 'students_name_idx']
  }
}

Table teachers {
  teacher_id bigint [pk, increment]
  user_id bigint [unique]
  department_id bigint
  employment_type varchar(30) [not null, default: 'full_time']
  teacher_code varchar(30) [not null, unique]
  first_name varchar(100) [not null]
  last_name varchar(100)
  gender varchar(20)
  phone varchar(30)
  email varchar(150) [unique]
  specialization varchar(150)
  hire_date date
  status varchar(30) [not null, default: 'active']
  address text
  created_at timestamp
  updated_at timestamp
}

Table enrollments {
  enrollment_id bigint [pk, increment]
  student_id bigint [not null]
  course_id bigint [not null]
  class_room_id bigint
  academic_year_id bigint
  semester_id bigint
  enrollment_date date
}
```

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```
status varchar(30) [not null, default: 'studying']
note text
created_at timestamp
updated_at timestamp

Indexes {
  (student_id, course_id, semester_id) [name: 'enrollments_student_course_semester_idx']
  (course_id, academic_year_id, semester_id, status) [name: 'enrollments_course_period_status_idx']
}

Table attendances {
  attendance_id bigint [pk, increment]
  student_id bigint [not null]
  class_room_id bigint
  attendance_date date [not null]
  status varchar(30) [not null, default: 'present']
  note text
  created_at timestamp
  updated_at timestamp

  Indexes {
    (student_id, attendance_date) [name: 'attendances_student_date_idx']
    (class_room_id, attendance_date) [name: 'attendances_class_date_idx']
  }
}

Table student_progresses {
  progress_id bigint [pk, increment]
  student_id bigint [not null]
  course_id bigint
  class_room_id bigint
  progress_date date
  progress_percent decimal(5,2) [not null, default: 0]
  score decimal(6,2)
  status varchar(30) [not null, default: 'in_progress']
  note text
  created_at timestamp
  updated_at timestamp

  Indexes {
    (student_id, course_id) [name: 'progress_student_course_idx']
    (course_id, progress_date) [name: 'progress_course_date_idx']
  }
}

Table certificates {
  certificate_id bigint [pk, increment]
  student_id bigint [not null]
  course_id bigint
  certificate_no varchar(50) [not null, unique]
  issued_date date
  status varchar(30) [not null, default: 'issued']
  note text
  created_at timestamp
  updated_at timestamp

  Indexes {
    (student_id, issued_date) [name: 'certificates_student_issued_idx']
    (course_id, issued_date) [name: 'certificates_course_issued_idx']
  }
}

Table course_assignments {
  course_assignment_id bigint [pk, increment]
  teacher_id bigint [not null]
  course_id bigint [not null]
  class_room_id bigint
  academic_year_id bigint
  assigned_date date
  status varchar(30) [not null, default: 'active']
  note text
  created_at timestamp
  updated_at timestamp

  Indexes {
    (course_id, teacher_id) [name: 'course_assignments_course_teacher_idx']
    (teacher_id, course_id) [name: 'course_assignments_teacher_course_idx']
  }
}
```

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```
}

Table teacher_schedules {
  teacher_schedule_id bigint [pk, increment]
  teacher_id bigint [not null]
  course_id bigint
  class_room_id bigint
  day_of_week varchar(20) [not null]
  start_time time [not null]
  end_time time [not null]
  room varchar(100)
  status varchar(30) [not null, default: 'active']
  note text
  created_at timestamp
  updated_at timestamp
}

Table teacher_performances {
  teacher_performance_id bigint [pk, increment]
  teacher_id bigint [not null]
  course_id bigint
  evaluation_date date [not null]
  teaching_score tinyint [not null, default: 0]
  attendance_score tinyint [not null, default: 0]
  student_feedback_score tinyint [not null, default: 0]
  rating varchar(30) [not null, default: 'good']
  comment text
  created_at timestamp
  updated_at timestamp
}

Table content_lessons {
  content_lesson_id bigint [pk, increment]
  course_id bigint [not null]
  module_number int [not null, default: 1]
  module_title varchar(255)
  title varchar(255) [not null]
  slug varchar(255) [not null, unique]
  content_type varchar(30) [not null, default: 'lesson']
  summary text
  body longtext
  external_url varchar(255)
  file_path varchar(255)
  video_url varchar(255)
  duration_minutes int
  position int [not null, default: 1]
  available_from datetime
  available_until datetime
  completion_required boolean [not null, default: false]
  visibility varchar(20) [not null, default: 'visible']
  max_score decimal(8,2)
  passing_score decimal(8,2)
  allow_comments boolean [not null, default: false]
  metadata json
  is_published boolean [not null, default: true]
  created_at timestamp
  updated_at timestamp
  deleted_at timestamp

  Indexes {
    (course_id, module_number, position)
    (content_type, visibility, is_published)
    (course_id, is_published, visibility, module_number, position) [name: 'lessons_course_visible_order_idx']
    (course_id, is_published, available_from, available_until) [name: 'lessons_course_schedule_idx']
  }
}

Table content_chapters {
  content_chapter_id bigint [pk, increment]
  content_lesson_id bigint [not null]
  title varchar(180) [not null]
  summary text
  content longtext
  sort_order int [not null, default: 0]
  is_published boolean [not null, default: false]
  created_at timestamp
  updated_at timestamp

  Indexes {
```

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```
(content_lesson_id, is_published, sort_order) [name: 'chapters_lesson_published_order_idx']
}
}

Table content_videos {
  content_video_id bigint [pk, increment]
  content_lesson_id bigint [not null]
  content_chapter_id bigint
  title varchar(180) [not null]
  description text
  video_path varchar(255)
  video_url varchar(255)
  duration_seconds int
  sort_order int [not null, default: 0]
  is_published boolean [not null, default: false]
  created_at timestamp
  updated_at timestamp

  Indexes {
    (content_lesson_id, is_published, sort_order) [name: 'videos_lesson_published_order_idx']
    (content_chapter_id, is_published, sort_order) [name: 'videos_chapter_published_order_idx']
  }
}

Table content_documents {
  content_document_id bigint [pk, increment]
  content_lesson_id bigint [not null]
  content_chapter_id bigint
  title varchar(180) [not null]
  description text
  file_path varchar(255) [not null]
  sort_order int [not null, default: 0]
  is_published boolean [not null, default: false]
  created_at timestamp
  updated_at timestamp

  Indexes {
    (content_lesson_id, is_published, sort_order) [name: 'documents_lesson_published_order_idx']
    (content_chapter_id, is_published, sort_order) [name: 'documents_chapter_published_order_idx']
  }
}

Table content_assignments {
  content_assignment_id bigint [pk, increment]
  content_lesson_id bigint [not null]
  content_chapter_id bigint
  title varchar(180) [not null]
  instructions longtext
  attachment_path varchar(255)
  due_at datetime
  max_score int [not null, default: 100]
  allow_late_submission boolean [not null, default: false]
  is_published boolean [not null, default: false]
  created_at timestamp
  updated_at timestamp

  Indexes {
    (content_lesson_id, is_published, due_at) [name: 'assignments_lesson_published_due_idx']
    (content_chapter_id, is_published, due_at) [name: 'assignments_chapter_published_due_idx']
  }
}

Table content_resources {
  content_resource_id bigint [pk, increment]
  content_lesson_id bigint [not null]
  content_chapter_id bigint
  title varchar(180) [not null]
  description text
  file_path varchar(255)
  external_url varchar(255)
  sort_order int [not null, default: 0]
  is_published boolean [not null, default: false]
  created_at timestamp
  updated_at timestamp

  Indexes {
    (content_lesson_id, is_published, sort_order) [name: 'resources_lesson_published_order_idx']
    (content_chapter_id, is_published, sort_order) [name: 'resources_chapter_published_order_idx']
  }
}
```

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```
}

Table quizzes {
  quiz_id bigint [pk, increment]
  content_lesson_id bigint
  title varchar(180) [not null]
  instructions longtext
  available_from datetime
  available_until datetime
  time_limit_minutes smallint
  max_attempts smallint [not null, default: 1]
  passing_score decimal(6,2) [not null, default: 50]
  is_published boolean [not null, default: false]
  created_at timestamp
  updated_at timestamp

  Indexes {
    (content_lesson_id, is_published, available_from) [name: 'quizzes_lesson_published_from_idx']
  }
}

Table question_banks {
  question_bank_id bigint [pk, increment]
  course_id bigint
  subject_id bigint
  title varchar(180) [not null]
  description text
  is_active boolean [not null, default: true]
  created_at timestamp
  updated_at timestamp
}

Table assessment_questions {
  assessment_question_id bigint [pk, increment]
  question_bank_id bigint
  content_lesson_id bigint
  question_text longtext [not null]
  question_type varchar(40) [not null, default: 'multiple_choice']
  points decimal(6,2) [not null, default: 1]
  correct_answer longtext
  explanation longtext
  is_active boolean [not null, default: true]
  created_at timestamp
  updated_at timestamp

  Indexes {
    (content_lesson_id, is_active) [name: 'questions_lesson_active_idx']
    (question_bank_id, is_active) [name: 'questions_bank_active_idx']
  }
}

Table question_options {
  question_option_id bigint [pk, increment]
  assessment_question_id bigint [not null]
  option_text varchar(500) [not null]
  is_correct boolean [not null, default: false]
  sort_order smallint [not null, default: 0]
  created_at timestamp
  updated_at timestamp

  Indexes {
    (assessment_question_id, sort_order) [name: 'options_question_order_idx']
  }
}

Table exams {
  exam_id bigint [pk, increment]
  course_id bigint
  subject_id bigint
  title varchar(180) [not null]
  description text
  exam_date date
  start_time time
  end_time time
  duration_minutes smallint
  total_score decimal(6,2) [not null, default: 100]
  passing_score decimal(6,2) [not null, default: 50]
  status varchar(30) [not null, default: 'draft']
  created_at timestamp
}
```

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```
    updated_at timestamp
}
```

```
Table exam_candidates {
    exam_candidate_id bigint [pk, increment]
    exam_id bigint [not null]
    student_id bigint [not null]
    seat_number varchar(50)
    status varchar(30) [not null, default: 'registered']
    created_at timestamp
    updated_at timestamp

    Indexes {
        (exam_id, student_id) [unique]
    }
}
```

```
Table exam_submissions {
    exam_submission_id bigint [pk, increment]
    exam_id bigint [not null]
    student_id bigint [not null]
    submitted_at datetime
    attempt_no smallint [not null, default: 1]
    answers json
    score decimal(6,2)
    status varchar(30) [not null, default: 'submitted']
    feedback longtext
    created_at timestamp
    updated_at timestamp
}
```

```
Table assessment_grades {
    assessment_grade_id bigint [pk, increment]
    student_id bigint [not null]
    exam_id bigint
    quiz_id bigint
    content_assignment_id bigint
    graded_by bigint
    score decimal(6,2) [not null, default: 0]
    max_score decimal(6,2) [not null, default: 100]
    grade varchar(20)
    graded_at datetime
    remarks text
    created_at timestamp
    updated_at timestamp

    Indexes {
        (student_id, graded_at) [name: 'grades_student_graded_idx']
        (student_id, exam_id) [name: 'grades_student_exam_idx']
        (student_id, quiz_id) [name: 'grades_student_quiz_idx']
        (student_id, content_assignment_id) [name: 'grades_student_assignment_idx']
    }
}
```

```
Table assessment_results {
    assessment_result_id bigint [pk, increment]
    student_id bigint [not null]
    exam_id bigint
    quiz_id bigint
    assessment_type varchar(30) [not null, default: 'exam']
    total_score decimal(6,2) [not null, default: 0]
    passed boolean [not null, default: false]
    rank int
    published_at datetime
    remarks text
    created_at timestamp
    updated_at timestamp

    Indexes {
        (student_id, quiz_id) [name: 'results_student_quiz_idx']
        (student_id, published_at) [name: 'results_student_published_idx']
    }
}
```

```
Table assignment_submissions {
    assignment_submission_id bigint [pk, increment]
    content_assignment_id bigint [not null]
    student_id bigint [not null]
    response longtext
}
```

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```
attachment_url varchar(1000)
submitted_at datetime
status varchar(30) [not null, default: 'submitted']
score decimal(6,2)
feedback longtext
created_at timestamp
updated_at timestamp

Indexes {
  (student_id, content_assignment_id) [name: 'submissions_student_assignment_idx']
  (student_id, submitted_at) [name: 'submissions_student_submitted_idx']
}
}

Table student_promotions {
  id bigint [pk, increment]
  student_id bigint [not null]
  from_department_id bigint
  from_year_id bigint
  from_semester_id bigint
  to_year_id bigint [not null]
  to_semester_id bigint [not null]
  promoted_at timestamp [not null]
  note text
  created_at timestamp
  updated_at timestamp
}

Table permissions {
  id bigint [pk, increment]
  name varchar(255) [not null]
  guard_name varchar(255) [not null]
  created_at timestamp
  updated_at timestamp

  Indexes {
    (name, guard_name) [unique]
  }
}

Table roles {
  id bigint [pk, increment]
  name varchar(255) [not null]
  guard_name varchar(255) [not null]
  created_at timestamp
  updated_at timestamp

  Indexes {
    (name, guard_name) [unique]
  }
}

Table model_has_permissions {
  permission_id bigint [not null]
  model_type varchar(255) [not null]
  model_id bigint [not null]

  Indexes {
    (permission_id, model_id, model_type) [pk, name: 'model_has_permissions_permission_model_type_primary']
    (model_id, model_type) [name: 'model_has_permissions_model_id_model_type_index']
  }
}

Table model_has_roles {
  role_id bigint [not null]
  model_type varchar(255) [not null]
  model_id bigint [not null]

  Indexes {
    (role_id, model_id, model_type) [pk, name: 'model_has_roles_role_model_type_primary']
    (model_id, model_type) [name: 'model_has_roles_model_id_model_type_index']
  }
}

Table role_has_permissions {
  permission_id bigint [not null]
  role_id bigint [not null]

  Indexes {
```


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```

(permission_id, role_id) [pk, name: 'role_has_permissions_permission_id_role_id_primary']
}
}

Ref: departments.faculty_id > faculties.faculty_id
Ref: semesters.academic_year_id > academic_years.academic_year_id
Ref: courses.course_category_id > course_categories.course_category_id
Ref: courses.department_id > departments.department_id
Ref: courses.academic_year_id > academic_years.academic_year_id
Ref: courses.semester_id > semesters.semester_id
Ref: subjects.course_id > courses.course_id
Ref: class_rooms.course_id > courses.course_id
Ref: class_rooms.academic_year_id > academic_years.academic_year_id
Ref: students.user_id - users.id
Ref: students.department_id > departments.department_id
Ref: students.academic_year_id > academic_years.academic_year_id
Ref: students.semester_id > semesters.semester_id
Ref: teachers.user_id - users.id
Ref: teachers.department_id > departments.department_id
Ref: enrollments.student_id > students.student_id
Ref: enrollments.course_id > courses.course_id
Ref: enrollments.class_room_id > class_rooms.class_room_id
Ref: enrollments.academic_year_id > academic_years.academic_year_id
Ref: enrollments.semester_id > semesters.semester_id
Ref: attendances.student_id > students.student_id
Ref: attendances.class_rooms.class_room_id > class_rooms.class_room_id
Ref: student_progresses.student_id > students.student_id
Ref: student_progresses.course_id > courses.course_id
Ref: student_progresses.class_room_id > class_rooms.class_room_id
Ref: certificates.student_id > students.student_id
Ref: certificates.course_id > courses.course_id
Ref: course_assignments.teacher_id > teachers.teacher_id
Ref: course_assignments.course_id > courses.course_id
Ref: course_assignments.class_room_id > class_rooms.class_room_id
Ref: course_assignments.academic_year_id > academic_years.academic_year_id
Ref: teacher_schedules.teacher_id > teachers.teacher_id
Ref: teacher_schedules.course_id > courses.course_id
Ref: teacher_schedules.class_room_id > class_rooms.class_room_id
Ref: teacher_performances.teacher_id > teachers.teacher_id
Ref: teacher_performances.course_id > courses.course_id
Ref: content_lessons.course_id > courses.course_id
Ref: content_chapters.content_lesson_id > content_lessons.content_lesson_id
Ref: content_videos.content_chapter_id > content_chapters.content_chapter_id
Ref: content_documents.content_lesson_id > content_lessons.content_lesson_id
Ref: content_documents.content_chapter_id > content_chapters.content_chapter_id
Ref: content_assignments.content_lesson_id > content_lessons.content_lesson_id
Ref: content_assignments.content_chapter_id > content_chapters.content_chapter_id
Ref: content_resources.content_lesson_id > content_lessons.content_lesson_id
Ref: content_resources.content_chapter_id > content_chapters.content_chapter_id
Ref: quizzes.content_lesson_id > content_lessons.content_lesson_id
Ref: question_banks.course_id > courses.course_id
Ref: question_banks.subject_id > subjects.subject_id
Ref: assessment_questions.question_bank_id > question_banks.question_bank_id
Ref: assessment_questions.content_lesson_id > content_lessons.content_lesson_id
Ref: question_options.assessment_question_id > assessment_questions.assessment_question_id
Ref: exams.course_id > courses.course_id
Ref: exams.subject_id > subjects.subject_id
Ref: exam_candidates.exam_id > exams.exam_id
Ref: exam_candidates.student_id > students.student_id
Ref: exam_submissions.exam_id > exams.exam_id
Ref: exam_submissions.student_id > students.student_id
Ref: assessment_grades.student_id > students.student_id
Ref: assessment_grades.exam_id > exams.exam_id
Ref: assessment_grades.quiz_id > quizzes.quiz_id
Ref: assessment_grades.content_assignment_id > content_assignments.content_assignment_id
Ref: assessment_grades.gradedy > users.id
Ref: assessment_results.student_id > students.student_id
Ref: assessment_results.exam_id > exams.exam_id
Ref: assessment_results.quiz_id > quizzes.quiz_id
Ref: assignment_submissions.content_assignment_id > content_assignments.content_assignment_id
Ref: assignment_submissions.student_id > students.student_id
Ref: student_promotions.student_id > students.student_id
Ref: student_promotions.from_department_id > departments.department_id
Ref: student_promotions.from_year_id > academic_years.academic_year_id
Ref: student_promotions.from_semester_id > semesters.semester_id
Ref: student_promotions.to_year_id > academic_years.academic_year_id
Ref: student_promotions.to_semester_id > semesters.semester_id
Ref: model_has_permissions.permission_id > permissions.id

```

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```
Ref: model_has_permissions.model_id > users.id
Ref: model_has_roles.role_id > roles.id
Ref: model_has_roles.model_id > users.id
Ref: role_has_permissions.permission_id > permissions.id
Ref: role_has_permissions.role_id > roles.id
```